Brian Wonjun Lee

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Education

University of Pennsylvania — Philadelphia, PA
MSE in Computer Graphics & Game Technology (CGGT), Submatriculation — *Expected May 2028*
BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars — *Expected May 2027*
Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

Experience

Software Engineer Intern, Aleph Lab (Y Combinator) — *June 2026 – Present · San Francisco (Remote)*

- Built and **shipped to production** a recency-segmented retention system (Supabase triggers, Deno edge functions, FCM) at a 4-engineer YC startup, with a treatment/holdout A/B that drove the product's **first measured win-back → paid conversion**
- Caught a win-back targeting flaw by validating against production data — repointing from child-role to cross-account activity for a **~2.7× gain in reachable users**
- Packaged a production AI agent as an **installable service** behind a typed SDK — 5 versioned wire contracts, **136 tests**; a fresh AI agent integrated a new environment in **0 fix-loops**
- **Root-caused a production auth outage** from a JWT signing-key migration and shipped the fix, restoring the guarded notification pipeline (401 → 200)

Software Engineer Intern, BitMango (Puzzle1 Studio) — *June 2025 – Aug 2025 · Pangyo, South Korea*

- Maintained a live portfolio of **50+ Unity mobile game titles**, improving UI/UX and core gameplay stability across deployments
- Resolved **100+ QA-reported bugs** by modifying Unity prefabs and C# scripts, shipping fixes in production releases
- Upgraded SDKs and platform modules **fleet-wide** to meet evolving app-store requirements and sustain production availability

Project Lead & Product Designer, Penn Spark — *Jan 2025 – Present · Philadelphia, PA*

- Led a **15+-person cross-functional team** of designers and developers to ship client-facing **0-to-1 products**
- Defined **end-to-end user flows**, interaction patterns, and reusable interface systems across multiple deliverables
- Shipped polished, **user-centered features** across semester-long client project cycles

Military Interpreter, Capital Defense Command (ROK Army) — *Dec 2022 – June 2024 · Seoul, South Korea*

- Provided **real-time English-Korean interpretation** in high-security, multi-agency environments for international stakeholders
- Translated **sensitive documents and briefings** with precision, enabling coordination across government and defense teams

Database Engineer Intern, IT-Farm — *June 2022 – Aug 2022 · Seongnam, South Korea*

- Designed and implemented a SQL database prototype managing **1M+ data entities** using Oracle and SQLite
- Processed and organized semiconductor datasets from **20+ client companies**, including Samsung, SK, and LG
- Wrote and **optimized SQL queries** to improve retrieval and management efficiency across large-scale datasets

Projects

Mini Minecraft | *C++, OpenGL, GLSL* — *Apr 2026*

- Engineered a **voxel engine from scratch**: 7 procedural biomes from layered Perlin/FBM/Voronoi noise with smoothstep blending
- Implemented **3D Perlin cave systems** with trilinear interpolation, fluid physics, and post-process tint overlays
- Built the render pipeline: **7×7 PCF soft shadows, screen-space reflections, vertex AO**, and a dynamic day-night cycle

Passenger | *Unreal Engine 5, HLSL, Maya* — *Mar 2026 – Apr 2026*

- Wrote 2 **HLSL post-process shaders** (anisotropic Kuwahara, cell shader) sampling GBuffer channels for stylized rendering
- **Rigged and animated a character** in Maya (IK limbs, spline-IK spine) driving an Unreal Animation Blueprint
- Rendered at 2560×1080 via Movie Render Pipeline with **hardware ray tracing** over Lumen GI, 64 TSR samples/frame

Mini Maya | *C++, OpenGL, Qt* — *Feb 2026 – Mar 2026*

- Implemented a **half-edge mesh data structure** enabling efficient traversal and topology-aware editing operations
- Built edge split, extrusion, and **Catmull–Clark subdivision** on an OpenGL/VBO pipeline with GLSL shaders

Technical Skills

Graphics & Rendering: OpenGL, GLSL, HLSL, shader programming, Unreal Engine 5, Lumen, real-time & offline rendering
3D & Tooling: Maya, Blender, Substance Painter, Qt, Movie Render Pipeline
Languages: C++, Python, TypeScript, JavaScript, Java, C#, GLSL/HLSL, SQL
Frameworks & Tools: React Three Fiber, Three.js, React, Node.js, Git, GitLab, VS Code